

The Collected Mathematical Papers of Arthur Cayley, Sc.D., F.R.S.

Vol. V

Pages 151–200

Cambridge: At the University Press. 1892

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334. (continued)

TABLES DES FORMES QUADRATIQUES BINAIRES (suite).

[Continuation from p. 150 of Table II. Positive non-square determinants $D = 2$ to $D = 99$; then Table III. Thirteen irregular negative determinants of the first thousand.]

The continuation of Table II is presented in the same form as before: for each determinant D , the number of classes, the characters, and the periods of reduced forms are tabulated. We resume at $D = 62$.

Table II (continued). Positive determinants $D = 62$ to $D = 99$.

D	Classes	Periods of the reduced forms
62	3	1, 0, -62; period of 1, 7, -13, 4, 2, 4, -13, 7, 1
63	3	1, 0, -63; 1, 7, -2, 7, 1; -1, 7, 2, 7, -1
65	1	1, 0, -65; period of 1, 8, -1, 8, 1
66	1	1, 0, -66; period of 1, 8, -2, 8, 1
67	1	1, 0, -67; period of 1, 8, -3, 7, 2, 7, -3, 8, 1
68	2	1, 0, -68; 2, 0, -17; periods of 1, 8, -4, 8, 1, -1, 8, 4, 8, -1
69	1	1, 0, -69; period of 1, 8, -5, 8, 4, 1, -5, 4, 8, 5, -8, 1
70	2	1, 0, -70; 2, 1, -23; periods of 1, 8, -6, 4, 9, 3, -6, 4, 1
71	1	1, 0, -71; period of 1, 8, -7, 7, 2, 7, -7, 8, 1
73	1	1, 0, -73; period of 1, 8, -9, 4, 9, 1, -8, 4, 9, -1
74	1	1, 0, -74; period of 1, 8, -10, 8, 1
77	1	1, 0, -77; period of 1, 8, -13, 4, 7, 7, -4, 13, 8, -1
78	2	1, 0, -78; 2, 0, -39; periods of 1, 8, -14, 2, 5, 9, -5, 2, 14, 8, -1
79	1	1, 0, -79; period of 1, 8, -15, 2, 9, 8, -7, 4, 11, 5, -3, 5, 11, 4, -7, 8, 9, 2, -15, 8, 1
80	2	1, 0, -80; 2, 0, -40; periods of 1, 8, -16, 8, 1
82	1	1, 0, -82; period of 1, 9, -1, 9, 1
83	1	1, 0, -83; period of 1, 9, -2, 9, 1
84	3	1, 0, -84; 2, 0, -21; 3, 0, -7; periods of 1, 9, -3, 9, 1
85	1	1, 0, -85; period of 1, 9, -4, 7, 3, 7, -4, 9, -1
86	1	1, 0, -86; period 1, 9, -5, 2, 13, 2, -5, 9, 1
87	2	1, 0, -87; 2, 1, -43; periods of 1, 9, -6, 9, 1
88	1	1, 0, -88; period of 1, 9, -7, 5, 9, 4, -7, 8, 3, -7, 4, 9, 5, -7, 9, 1
89	1	1, 0, -89; period of 1, 9, -8, 8, 3, 8, -8, 9, 1
90	3	1, 0, -90; 2, 0, -45; 3, 0, -30; periods of 1, 9, -9, 9, 1
91	2	1, 0, -91; 2, 1, -45; periods of 1, 9, -10, 3, 9, 4, -7, 3, 12, 5, -3, 5, 12, 3, -7, 4, 9, 3, -10, 9, 1
92	1	1, 0, -92; period of 1, 9, -11, 9, 1
93	1	1, 0, -93; period of 1, 9, -12, 4, 7, 8, -3, 12, 1, -12, 3, 8, -7, 4, 12, 9, -1
94	1	1, 0, -94; period of 1, 9, -13, 5, 9, 7, -7, 4, 15, 2, -15, 4, 7, 7, -9, 5, 13, 9, 1
95	2	1, 0, -95; 2, 1, -47; periods of 1, 9, -14, 4, 11, 5, -14, 9, 1
96	2	1, 0, -96; 2, 0, -48; periods of 1, 9, -15, 5, 7, 8, -5, 15, 9, 1
97	1	1, 0, -97; period of 1, 9, -16, 2, 15, 3, -13, 4, 12, 5, -12, 4, 13, 3, -15, 2, 16, 9, 1
98	2	1, 0, -98; 2, 0, -49; periods of 1, 9, -17, 9, 1
99	2	1, 0, -99; 2, 1, -49; periods of 1, 9, -18, 9, 1

Table III. The thirteen irregular negative determinants of the first thousand.

For each such determinant the table exhibits the reduced forms in their proper classes, together with the characters and the auxiliary symbols Cp . The irregular determinants of the first thousand are

$$D = -576, -580, -820, -884, -900, -1(24)^2, -145(3)^2, -205(1)^2, \\ -243(9)^2, -307(1)^2, -339(1)^2, -459(9)^2, -675(15)^2.$$

(The notation $-D(k)^2$ indicates an irregular determinant, k being the parameter associated with the irregularity.)

For example, for $D = -576$ the reduced forms are

$$(1, 0, 576), (5, 2, 116), (8, 4, 73), (9, 3, 65), (13, 5, 46), (16, 4, 37), (16, 8, 40), (17, 3, 34), \dots$$

with classes distinguished by the characters $\pm$ in the eight columns of generic characters, and by the auxiliary symbols Cp in the last column.

For $D = -580$ the reduced forms include

$$(1, 0, 580), (4, 2, 146), (5, 0, 116), (8, 2, 73), (10, 0, 58), (13, 3, 45), \dots$$

For $D = -820$ the forms include

$$(1, 0, 820), (4, 2, 206), (5, 0, 164), (10, 0, 82), (11, 1, 75), (16, 6, 52), (20, 8, 44), \dots$$

For $D = -884$ the forms include

$$(1, 0, 884), (5, 3, 177), (9, 4, 100), (13, 3, 69), (17, 7, 53), (20, 2, 45), (21, 7, 43), \dots$$

For $D = -900$ the forms include

$$(1, 0, 900), (9, 0, 100), (13, 0, 75), (17, 4, 53), (25, 0, 36), (29, 2, 32), \dots$$

For $D = -1(24)^2 (= -576)$ the period of the irregular class is exhibited, with the auxiliary indications.

For $D = -145(3)^2$ the table gives

$$(1, 0, 1305), (5, 0, 261), (9, 4, 145), (10, 5, 131), (17, 8, 77), (18, 7, 73), (25, 10, 53), \dots$$

For $D = -205(1)^2$ the table gives

$$(1, 0, 820), (4, 1, 205), (5, 0, 164), (16, 7, 52), (20, 5, 42), \dots$$

For $D = -243(9)^2$ the table gives

$$(1, 0, 243), (3, 0, 81), (7, 2, 35), (9, 0, 27), (11, 4, 23), (13, 1, 19), \dots$$

For $D = -307(1)^2$ the table gives

$$(1, 0, 307), (4, 1, 77), (7, 1, 44), (11, 1, 28), (13, 3, 24), (17, 4, 19), \dots$$

For $D = -339(1)^2$ the table gives

$$(1, 0, 339), (4, 1, 85), (5, 1, 68), (13, 5, 28), (15, 6, 25), (17, 1, 20), \dots$$

For $D = -459(9)^2$ the table gives

$$(1, 0, 459), (4, 1, 115), (9, 0, 51), (13, 3, 36), (17, 5, 28), (19, 4, 25), \dots$$

For $D = -675(15)^2$ the table gives

$$(1, 0, 675), (3, 0, 225), (4, 1, 169), (11, 1, 62), (19, 8, 37), (25, 5, 28), \dots$$

[The full tabulation of these thirteen irregular determinants occupies the remainder of pages 154–156 of the original. We refer the reader to the printed source for the complete listing.]

336.

NOTE SUR L'ÉLIMINATION.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LX. (1861), pp. 373–374.]

Soient $U = (a, \dots \check{x}, y)^p$, $V = (b, \dots \check{x}, y)^q$ des fonctions homogènes quelconques des degrés m et n respectivement. Dénotons par $(x, y)^p$ la suite entière ou seulement une partie de la suite de termes $x^p, x^{p-1}y, \dots, y^p$, et en prenant $\theta \leq m \leq p$, formons le déterminant

$$|(x, y)^{p-m}U, (x, y)^{p-n}V|.$$

Cette notation signifie qu'on a supposé les suites $(x, y)^{p-m}U, (x, y)^{p-n}V$ composées respectivement de p et de q termes et qu'on posant $p + q = s$ on forme le déterminant

$$\begin{vmatrix} (x_1, y_1)^{p-m}U_1 & (x_1, y_1)^{p-n}V_1 \\ \vdots & \vdots \\ (x_s, y_s)^{p-m}U_s & (x_s, y_s)^{p-n}V_s \end{vmatrix},$$

dans lequel les différentes lignes (chacune composée de s termes) sont ce que deviennent $(x, y)^{p-m}U, (x, y)^{p-n}V$, lorsqu'on y substitue $(x_1, y_1), (x_2, y_2), \dots, (x_s, y_s)$ successivement au lieu de (x, y) .

Le déterminant que je viens de définir est divisible par le déterminant

$$|(x, y)^{s-1}|,$$

notation qui est équivalente à:

$$\begin{vmatrix} (x_1, y_1)^{s-1} \\ \vdots \\ (x_s, y_s)^{s-1} \end{vmatrix},$$

et dans laquelle $(x, y)^{s-1}$ dénote la suite entière des termes $x^{s-1}, x^{s-2}y, \dots, y^{s-1}$. Nous obtenons ainsi une équation

$$\frac{|(x, y)^{p-m}U, (x, y)^{p-n}V|}{|(x, y)^{s-1}|} = (a, \dots)^p (b, \dots)^q (x_1, y_1)^{\alpha+\beta-1} \dots (x_s, y_s)^{\alpha+\beta-1},$$

c.-à-d. que le quotient est du degré p par rapport aux coefficients $(a, \dots)$, du degré q par rapport aux coefficients $(b, \dots)$, et du degré $\theta - s + 1$ par rapport à chaque système de variables (x_i, y_i) . Or en supposant $U_i = 0, V_i = 0$, on obtient

$$0 = (a, \dots)^p (b, \dots)^q (x_1, y_1)^{\theta-s+1} \dots (x_s, y_s)^{\theta-s+1},$$

équation qui subsiste quelles que soient les valeurs des variables $(x_1, y_1), \dots, (x_s, y_s)$; cela donne une suite de $\theta - s + 2(s - 1)$ équations chacune de la forme

$$0 = (a, \dots)^p (b, \dots)^q (x_i, y_i)^{\theta-s+1}.$$

En considérant un système quelconque de $\theta - s + 2$ de ces équations, pour en éliminer tous les termes de $(x_i, y_i)^{\theta-s+1}$, on obtiendra ou bien l'équation identique $0 = 0$, ou bien une équation de la forme

$$F = (a, \dots)^{p-\theta-s+2} (b, \dots)^{q-\theta-s+2} = 0$$

où F sera un déterminant de l'ordre $\theta - s + 2$, chaque terme étant de la forme $(a, \dots)^p (b, \dots)^q$.

Cela posé il est évident que F contiendra comme facteur la fonction

$$\square = (a, \dots)^n (b, \dots)^m$$

qui est le résultant des deux équations $U = 0$, $V = 0$. En particulier, on aura les deux cas que voici :

1°. Soit $\theta = m + n - 1$, et supposons que $(x, y)^{p-m}U$, $(x, y)^{p-n}V$, dénotent les suites entières

$$x^{m-1}U, x^{m-2}yU, \dots, y^{m-1}U; \quad x^{n-1}V, x^{n-2}yV, \dots, y^{n-1}V,$$

nous aurons $p = n$, $q = m$, $s = m + n$, $\theta - s + 2 = 1$, et de là

$$F = (a, \dots)^n (b, \dots)^m,$$

donc $F = \square$. On voit sans peine que l'on obtient de cette manière le résultant $\square$ sous la forme d'un déterminant de l'ordre $m + n$, le même que l'on obtient en éliminant les termes de $(x, y)^{m+n-1}$ entre les équations $(x, y)^{n-1}U = 0$, $(x, y)^{m-1}V = 0$, c.-à-d. par la méthode abrégée de Bézout.

2°. En supposant $m \leq n$, on peut prendre $\theta = m$, ne pas donner pour $(x, y)^{p-m}U$ le seul terme U . Réduisons aussi $(x, y)^{p-n}V$ au seul terme $x^{p-n}y^{p-n} \dots V$ (x désignant un nombre entier arbitraire entre 0 et $m - n$), c.-à-d. au terme $x^{p-n}y^{m-n}V$ ou $y^{m-n}V$ dans le cas des deux valeurs extrêmes de α . On a ainsi $p = 1$, $q = 1$, $s = 2$, et de là

$$F = (a, \dots)^n (b, \dots)^p.$$

Donc $F = (a, \dots)^{m-n} \square$, c.-à-d. que l'on obtient le résultant $\square$ affecté d'un facteur $(a, \dots)^{m-n}$ qui ne contient que les coefficients de U , et qui est de l'ordre $m - n$ par rapport à ces coefficients. L'expression de ce facteur peut être trouvée assez facilement. Dans le cas du terme $x^{p-n}y^{m-n}V$, c.-à-d. pour $\alpha = m - n$, le facteur sera k^{m-n} (k désignant le dernier coefficient de la suite $(a, \dots)$), et dans le cas du terme $y^{m-n}V$, c.-à-d. pour $\alpha = 0$, le facteur sera a^{m-n} . Mais en supposant $m = n$ on a tout simplement $F = \square$, c.-à-d. que l'on obtient le résultant sans facteur étranger. C'est sous cette dernière forme que j'ai présenté la méthode abrégée de Bézout dans le tome LIII. p. 366 (1857) de ce Journal, [250].

Londres, 17 Décembre, 1861.

337.

NOTE SUR LA RÉALITÉ DES RACINES D'UNE ÉQUATION QUADRATIQUE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LXI. (1863), pp. 367-368.]

À propos du mémoire que vient de publier M. Hesse (*voir ce Journal* t. LX, p. 305) je remarque que si l'on a l'une ou l'autre des deux formes

$$(a, b, c, f, g, h \text{ } \text{ } x, y, z)^2, \quad (a', b', c', f', g', h' \text{ } \text{ } x, y, z)^2$$

est une forme *définie* (forme qui conserve toujours le même signe pour des valeurs réelles quelconques des variables), l'équation suivante en λ :

$$\begin{vmatrix} a - \lambda a' & h - \lambda h' & g - \lambda g' & x \\ h - \lambda h' & b - \lambda b' & f - \lambda f' & y \\ g - \lambda g' & f - \lambda f' & c - \lambda c' & z \\ x & y & z & \end{vmatrix} = 0$$

aura ses deux racines réelles. En écrivant

$$(A, \dots \check{x}, y, z)^2 = \lambda(A, \dots)(x', \dots) - \lambda^2(A', \dots)(x', \dots)^2, \quad (a', b', c', f', g', h' \check{x}, y, z)^2 = 0,$$

$$B = ca - g^2, \dots$$

de manière que $(A, B, C, F, G, H \check{x} \bar{y}, \bar{z})$ dénote la forme adjointe (ou réciproque) de $(a, b, c, f, g, h \check{x}, y, z)$, cette équation prend la forme

$$(A, \dots \check{x}, y, z)^2 (A', \dots \check{x}, y, z)^2 - [(A, \dots \check{x}, y, z)^2]^2 = 0,$$

et les racines étant réelles, on doit avoir

$$\square = -4(A, \dots \check{x}, y, z)^2 (A', \dots \check{x}, y, z)^2 + [(A, \dots \check{x}, y, z)^2]^2 = +.$$

Or pour démontrer directement cette proposition, il n'est pas ce me semble possible d'exprimer $\square$ comme une somme de carrés; on a besoin de considérer une forme plus générale, savoir une somme de carrés multipliés chacun par un coefficient littéral positif. Par exemple, en ne faisant attention qu'au coefficient de x^4 , on doit avoir

$$\square_x = -4(bc - f^2)(b'c' - f'^2) + (bc' + b'c - 2ff')^2 = +.$$

Pour en faire la démonstration, on peut exprimer $\square_x$ sous la forme

$$\square_x = (bc' - b'c)^2 + 4(bf' - b'f)(cf' - c'f),$$

ce qui donne

$$bc\square_x = (bc - f^2)(bc' - b'c)^2 + (bc' - cf')(bc' - cf) + c(bf' - b'f)^2.$$

En effet, en y substituant la seconde expression de $\square_x$, on a l'identité

$$4bc(bf' - b'f)(cf' - c'f) = -f^2(bc' - b'c)^2 + (bcf' - c'f) + c(bf' - b'f)^2$$

et l'expression pour $bc\square_x$ est ainsi démontrée. Mais en supposant que $(a, b, c, f, g, h)(.,.)^2$ soit une forme définie, on a $bc - f^2 = +$, donc enfin $\square_x = +$. Il serait assez intéressant de trouver une démonstration pareille pour l'expression générale de $\square$.

Londres, 22 Octobre 1862.

338.

NOUVELLES RECHERCHES SUR L'ÉLIMINATION ET LA THÉORIE DES COURBES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LXIII. (1864), pp. 34–39.]

Dans le problème de l'élimination, on cherche la relation qui doit exister entre les coefficients d'un système de fonctions pour que quelque circonstance particulière (ou singularité) puisse avoir lieu; par exemple, pour que deux équations puissent avoir une racine commune, ou (comme application géométrique) pour qu'une courbe puisse avoir un point double. En prenant les coefficients comme données, tant la relation cherchée que la singularité qu'elle implique n'ont qu'une certaine existence *hypothétique*. Mais on peut transformer la question en supposant que les coefficients d'une ou plusieurs des fonctions soient de la forme $a = \lambda a' + \mu a''$, $b = \lambda b' + \mu b''$, ..., où a' , b' , ..., a'' , b'' , ... sont des coefficients donnés, mais λ , μ des quantités arbitraires. On peut alors disposer de sorte que la singularité dont il s'agit existe actuellement, en déterminant, au moyen de la relation donnée par l'élimination, la valeur du rapport $\lambda : \mu$. Ces substitutions $a = \lambda a' + \mu a''$, $b = \lambda b' + \mu b''$, ... changent la fonction U à laquelle se rapportent les coefficients a , b , ... en $U = \lambda U' + \mu U''$, où U' , U'' sont des fonctions semblables à U , mais avec les coefficients a' , b' , ... ou a'' , b'' , ..., au lieu de a , b , ...; et en servant d'une expression usitée, on peut dire que la fonction U est en involution avec U' , U'' ; et de même en géométrie que la courbe $U = 0$ est en involution avec les courbes $U' = 0$, $U'' = 0$; au reste, pour les courbes cela veut dire que les trois courbes se rencontrent dans les mêmes points.

On conçoit comment cette manière d'envisager le problème peut conduire à une interprétation géométrique de résultats qui n'avaient auparavant qu'une signification analytique. Considérons par exemple la proposition suivante, "le discriminant d'une fonction quadratique à trois variables est du degré 3 par rapport aux coefficients," ou ce qui est la même chose, "la fonction qui égale à zéro exprime que la conique

$$U = 0$$

ait un point double ou se réduise à une paire de droites (c'est du degré 3 par rapport aux coefficients, c'est à dire qu'en mettant pour chaque coefficient a , b , ... une telle fonction $\lambda U' + \mu U''$, et puis en égalant à zéro le théorème géométrique que voici: "Dans le système des coniques $\lambda U' + \mu U'' = 0$ en involution avec les coniques données $U' = 0$, $U'' = 0$, il y a 3 coniques à point double (c'est-à-dire, trois paires de droites) et le théorème sur le degré du discriminant des fonctions $\lambda U' + \mu U'' = 0$." On considère en question si on considérerait, on peut dire que la fonction U est en involution avec les courbes $U' = 0$, $U'' = 0$.

La fonction $U = (A, \dots \check{x}, y, z)^p$ étant telle que ne contient que des coefficients donnés et un nombre β de points de rebroussement, on a un discriminant spécial est $3(n-1)^2 - 7a - 11\beta$.

Sous la désignation de "discriminant spécial" j'entends la fonction laquelle égalée à zéro donne la condition pour que la courbe $U = 0$ ait un point double de plus. Il convient de remarquer par rapport à cette expression que le discriminant de la fonction générale du $n^{\text{ième}}$ ordre, en y substituant, au lieu des valeurs générales, les coefficients de la fonction U dont il s'agit, ne donne aucunement le discriminant spécial de la fonction U . En parlant tout simplement de l'ordre du discriminant spécial, j'ai voulu désigner l'ordre auquel s'élèvent par rapport à des coefficients arbitraires les éléments a , b , ... lorsqu'on suppose dans la fonction générale; et la formule énoncée ci-dessus donne en effet le rebroussement pour exprimer dans la forme signalée, c'est-à-dire linéairement par rapport à des coefficients absolument arbitraires comme éléments a , b , ...; proposition qui peut être démontrée sans difficulté.

Considérons en effet l'équation générale $U = (A, \dots \checkmark x, y, z)^p = 0$ les coefficients $A, \dots$ sont tous arbitraires; dans le cas d'un point double supposons que les coordonnées de ce point soit comme on doit dire, et les coordonnées des tangentes en ce point doivent satisfaire à chaque condition; on a en conséquence un système équivalent à trois conditions de cinq variables dans la direction de la tangente en ce point; donc la fonction U doit avoir un point double et il s'élève par les valeurs cherchées donc connues $U = 0, U' = 0$, quel est le nombre des points dans tout point double 'a tout point double' ou, ce qui est la même chose, le nombre des points dans tout point double 'a tout point double' du système. On considère la question si non désirer dans le système; forme, ces coordonnées données $U = 0, U' = 0$, quel est le nombre des courbes parmi les courbes données par les équations $\lambda U' + \mu U'' = 0$ en involution avec les deux courbes données. En considérant la question si on considérerait dans le système; forme, ces coordonnées données $U = 0, U' = 0$, quel est le nombre des courbes parmi les courbes données par les équations $\lambda U' + \mu U'' = 0$.

[The remainder of paper 338 (pp. 165–167) continues the same analytical investigation: deduction of formulae involving the discriminantal numbers, the orders of contact, and the number of points of the various kinds attaching to a curve in involution with two given curves. We refer the reader to the printed source for the residual algebraic computations.]

Londres, 22 Mai 1863.

339.

ON SKEW SURFACES, OTHERWISE SCROLLS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLIII. (for the year 1863) pp. 453–483. Received February 3,—Read March 5, 1863.]

It may be convenient to mention at the outset that, in the paper “On the Theory of Skew Surfaces”¹, I pointed out that upon any skew surface of the order n there is a singular (or nodal) curve meeting each generating line in $(n - 2)$ points, and that the class of the circumscribed cone (or, what is the same thing, the class of the curve, on the consideration of the lines of the surface as it meets an arbitrary plane) is equal to or less than n ; and the further development, by Dr Salmon, of this theory is somewhat more particularly as regards the Degree of a Surface reciprocal to a given one²: “On the Degree of a Surface reciprocal to a given one”³, where certain minor limits are given for the Degree of a Surface reciprocal to a given one (*), where certain minor limits are given for the Degree of a Surface reciprocal to a given one. The theory is somewhat further developed in Dr Salmon’s memoir “On the Degree of a Surface reciprocal to a given one”⁴, of a Surface reciprocal to a given one (*), where certain minor limits are given for the Degree of a Surface reciprocal to a given one. And in the equations $(\alpha, \dots \checkmark x, y, z, w)^n = 0$, $(\alpha', \dots \checkmark x, y, z, w)^n = 0, \dots$ are any linear functions of the coordinates, and t is an arbitrary parameter. And the same theories are reproduced in the “Treatise on the Analytic Geometry of Three Dimensions”⁵.

¹ *Cambridge and Dublin Math. Journ.* vol. vii. pp. 171–173 (1852), [108].

² *Trans. Royal Irish Acad.* vol. xxiii. pp. 461–488 (read 1855).

³ Dublin, 1862. [Ed. 4, 1882.]

⁴ Same reference as above.

⁵ Dublin, 1862. [Ed. 4, 1882.]

I will, though it is less closely connected with the subject of the present memoir, refer to a paper by M. Chasles, "Description des courbes à double courbure de tous les ordres sur les surfaces réglées du troisième et du quatrième ordre"(*).

The present memoir (in the composition of which I have been assisted by a correspondence with Dr Salmon) contains a further development of the theory of the connexion of the curves m, n, p respectively; 2nd, the surface generated by a line which meets a curve $S(m, n, p)$, and two lines which meet a given curve m, n, p respectively; 2nd, the surface generated by a line which meets a given curve m, n, p respectively; namely, that of three given curves m, n, p respectively; 2nd, the surface generated by a line which meets a given curve m, n, p respectively; and in the investigation of these I have been led somewhat to vary the notation of the previous memoir, but the alteration is one of detail only, the equations being substantially the same.

The present memoir (in the composition of which I have been assisted by a correspondence with Dr Salmon) contains a further development of the theory, viz. 1st, the surface generated by a line meeting three given curves; 2nd, the surface generated by a line meeting a given curve twice, and a given line once; 3rd, the surface generated by a line meeting a given curve in three points; and in the investigation of these I have been led somewhat to vary the notation of the previous memoir, but the alteration is one of detail only, the equations being substantially the same.

Notation and Table of Results, Articles 1 to 10.

1. In the present memoir a letter such as m denotes the order of a curve in space. It is for the most part assumed that a correspondence with Dr Salmon contains a further development of the theory of the connexion of the curves containing a further development of the theory. Thus $G(m, n, p, q)$ denotes the order of any one of the surfaces, that is, the surface generated by a line meeting a given line and three curves m, n, p, q respectively; and similarly $G(m, n)$ denotes the order of the surface generated by a line meeting a given line twice and two curves m, n respectively; 2nd, the surface generated by a line meeting a given curve in two points, and the surface generated by a line meeting a given curve in two points are surfaces of the same kind, the simplification being merely in the notation.

2. The use of the combinations (m, n, p, q) , (m^2, n, p) , &c. hardly requires explanation; it may however be noticed that the order of the surface $G(m, n, p)$ denotes the number of intersections of two non-coincident generating lines of the surface. This gives for the cases in question:

$$\Pi(m, n, p) = G(m, n, p), \dots$$

and the like manner, if $S(m)$ stands for the order of the scroll generated by a line meeting a given curve m in 4 points, &c., $(m, n, p) = 2mnp$, $(m^2, n) = m^2n$, $G(m, n, p) = G(m, n, p)$ denotes the order of the surface, (m, n, p) , and the number of nodal lines of which is $\frac{1}{2}(m + n - 1)(n - 1)$, the order denotes the order of the surface. The same will be employed throughout the memoir; and the order of the surface, the order being merely the order of the scroll.

3. The letters G, ND, NG, NR, NT (read Generators, Scroll, Nodal Director, Nodal Generator, and Nodal Total, are, in the simplest case in which there is a single curve of intersection, the orders m, n , and a generating line meets each of these in a single point.

$$NT(m, n, p) = ND(m, n, p) + NG(m, n, p),$$

and similarly for the scrolls $S(m^2, n)$ and $S(m^3)$.

4. $G(m, n, p, q)$. — Taking any point on the curve m , this is the vertex of two cones passing through the curves n, p respectively; the cones are of the orders n, p , respectively, and their

intersection therefore in np lines, which are the generating lines (through the assumed point) of the surface. Hence we have

$$G(m, n, p, q) = 2mnpq.$$

The line meeting these may be drawn through any point of the curve m , and it will sweep out the entire surface in the order m .

5. $G(m^2, n, p)$. — Taking any point on the curve n , the two chords through this point of the curve m form, together with the curve p , a triangle, the lines from the point to the curve p , of the order p giving in all $p \cdot \frac{1}{2}m(m-1)$, and the same is the order of the scroll through this is $\frac{1}{2}m(m-1)$ chord, that is, the order of the scroll through the point on n . The like investigation, taking points on the curve p , gives the order of the surface generated by a line which meets a given curve m in two points, and the curve p once, is

$$G(m^2, n, p) = \frac{1}{2}p \cdot m(m-1) \cdot n + (\text{like terms with } n, p \text{ interchanged}).$$

6. $G(m^3, p)$. — Through any point of the curve p , the three sides of the triangle formed by the three chords of the curve m , the order p giving therefore $p \cdot \frac{1}{6}m(m-1)(m-2)$. The order of the surface generated by a line meeting a given curve m in three points and a given curve p once is

$$G(m^3, p) = \frac{1}{6}p \cdot m(m-1)(m-2).$$

7. $NG(m, n, p, q)$. — $NG(m, n, p)$. — $NG(m^2, n) = NG(m^2)$. — $NG(m^3)$. — We have here the case in which a line passes through two points of the curve m ; and consider any one of the surfaces (G surfaces); the line in question is a generating line of the surface, but is also a nodal line: when this is the case, the line meets the surface in $4(n-1)^2$ points, and so on.

8. $NR(m, n, p, q)$. — $NR(m, n, p)$. — $NR(m^2, n)$. — $NR(m^3)$. — Here, the nodal residue is the intersection of the surface with another surface, formed by considering the assumed generating line as the intersection of two surfaces; from the order of the curve of intersection thus formed, the order of the generating line itself being subtracted, we obtain the order of the nodal residue.

9. $NT(m, n, p, q)$. — $NT(m, n, p)$. — $NT(m^2, n)$. — $NT(m^3)$. — These are the total nodal curves, viz. the union of the nodal director and the nodal generator; we have

$$NT(m, n, p) = ND(m, n, p) + NG(m, n, p) + NR(m, n, p),$$

and similarly for the scrolls $S(m^2, n)$ and $S(m^3)$.

10. I remark that the formulae are best exhibited in an order different from that in which they are in the sequel obtained, viz. I collect them in the following

Table.

$$\begin{aligned} G(m, n, p, q) &= 2mnpq, \\ G(m^2, n, p) &= np([m]_2 + M), \\ G(m^3, p) &= p([m]_3 + N \cdot \frac{1}{2}[m]_2 \cdot \frac{1}{2} + MN), \\ G(m^4) &= [m]_4 + N \cdot [m]_2 \cdot \frac{1}{2} - 2[m]_2 + M^2 \cdot \frac{1}{2}, \\ S(m, n, p) &= 2mnp, \\ ND(m, n, p) &= \frac{1}{2}mnp(mn + mp + np - 3), \\ NG(m, n, p) &= \frac{1}{2}mnp(4mnp - mn - mp - np + 3) - Mnp - Nmp - Pmn, \\ NR(m, n, p) &= \frac{1}{2}mnp(4mnp - 7) - 5 + Mnp + Nmp + Pmn, \\ NT(m, n, p) &= 2mnp(mnp - 1) + Mnp + Nmp + Pmn, \end{aligned}$$

together with the further entries

$$\begin{aligned} S(1, 1, m) &= 2m, \\ ND(1, 1, m) &= [m]_2, \\ NG(1, 1, m) &= [m]_2 + M, \\ NR(1, 1, m) &= 0, \\ NT(1, 1, m) &= 2[m]_2 + M, \end{aligned}$$

and

$$\begin{aligned} S(1, m, n) &= 2mn, \\ ND(1, m, n) &= \frac{1}{2} mn (mn + m + n - 3), \\ NG(1, m, n) &= mn (m + n - 2) + Mn + Nm, \\ NR(1, m, n) &= \frac{1}{2} mn (mn - m - n - 3) - Mn - Nm, \\ NT(1, m, n) &= \frac{1}{2} mn (mn + 3m + 3n - 5); \end{aligned}$$

moreover

$$\begin{aligned} S(m^2, n) &= n ([m]_2 + M), \\ ND(m^2, n) &= n \left(\frac{1}{2} [m]_2^2 + [m]_2 \cdot M + \frac{1}{2} [m]_2 + \frac{1}{2} M^2 - \frac{1}{2} \right), \\ NG(m^2, n) &= n \left(2 [m]_2^2 + [m]_2 \cdot M + \frac{3}{2} [m]_2 - 2 \right) + Mn, \\ NR(m^2, n) &= 0, \\ NT(m^2, n) &= n \left(\frac{5}{2} [m]_2^2 + 2 [m]_2 \cdot M + 2 [m]_2 + \frac{1}{2} M^2 + \frac{1}{2} \right) - Mn. \end{aligned}$$

The ND formulae, Articles 11 to 16.

11. $ND(m, n, p)$.— Taking any point on the curve m , this is the vertex of two cones passing through the curves n, p respectively; the cones are of the orders n, p , respectively, and their intersection therefore in np lines, the generating lines through the assumed point of the scroll $S(m, n, p)$. Hence

$$ND(m, n, p) = \frac{1}{2} mnp (mn + mp + np - 3).$$

12. $ND(m^2, n)$.— Taking first a point on the curve m , this is the vertex of a cone of the order $m - 1$ through the curve m , that is, of the cone of the order $m - 1$ through the remaining $m - 1$ points of intersection with the curve. Taking next a point on the curve n , this is the vertex of a cone passing through the curves $m, \frac{1}{2} [m]_2 + M$ double lines, which are the generating lines (through the assumed point) of the scroll $S(m^2, n)$. Hence we have

$$ND(m^2, n) = n \cdot \left[\frac{1}{2} [m]_2^2 + [m]_2 \cdot M + \frac{1}{2} [m]_2 + \frac{1}{2} M^2 - \frac{1}{2} \right].$$

13. $ND(m^3)$.— Taking any point on the curve m , in the vertex of a cone of the order $m - 1$ through the curve m at $m - 2$ other points, the cone is of the order $\frac{1}{2} [m]_2 - m + 2 + M$ double lines, which are the generating lines on the scroll $S(m^3)$. Hence we have

$$ND(m^3) = m \left[\frac{1}{2} [m]_2^2 - m + 2 + M \right] = m \left[\frac{1}{2} [m]_2^2 + \frac{1}{2} [m]_2 + M [m]_2 + \frac{1}{2} M^2 \cdot \frac{1}{2} m \right].$$

Preparatory remarks in regard to the G formulae, the hypertrioidal singularities of a curve in space, Articles 14 to 22.

14. It is to be remarked that the generating line of any one of the scrolls $S(m, n, p)$, $S(m^2, n)$, $S(m^3)$ satisfies three conditions; and that it cannot in anywise happen that one of these conditions could be in the other two. Thus, for instance, in the generating line of the scroll $S(m, n, p)$ the line meeting the two curves m, n which are met by the generating line in the same point; that

is, in fact have, as will be developed somewhat more particularly as regards the line $G(m, n, p)$ in question.

15. The foregoing investigation is not very satisfactory, but I confirm it by considering the case of two plane curves, orders m and n , and classes μ and ν , respectively. The tangents of the two curves can, it is clear, only meet on the line of intersection of the planes; in fact as follows: from any point of the line of intersection draw a tangent to each of the curves; the construction of the Torse is in fact as follows: from any point of the line of intersection draw a tangent to each of the curves; this is therefore $\mu\nu$ in number.

16. Given a curve m , then (as in fact mentioned in the investigation for $ND(m^2)$) through any point not whatever of the curve there may be drawn

$$(h - m + 2) ([m]_2 + m - 2) \text{ lines,}$$

meeting the curve in two other points, or say $[m]_2 + m - 2$ lines through three points, which can be drawn a line through three points; the number of these is in general $\frac{1}{6}(m-2)([m]_2 + m - 2)$.

17. But the curve m may be such that through every point of the curve there passes a line through four points. In fact, in general every line meets the curve in the points where it is met by it, and following from these to draw the curve m through them, the curve passes through these is the only place; in fact in such a case we cannot draw a chord through these points; in general every chord drawn meets the curve in two points only; but if a chord cannot be drawn through five points, we cannot, in general, draw a chord through six points; in general every chord drawn meets the curve in two points only; but if a chord cannot be drawn through five points, we cannot, in general, draw a chord through six points.

18. It is to be noticed, moreover, that if we take an arbitrary point of the curve, we may have the points $(\alpha, \beta, \gamma, \delta)$ on the curve. The like remarks apply to the lines $G(m^2, n, p)$, $G(m^2, n)$, and $G(m^3)$; but I will develop them somewhat more particularly as regards the line $G(m^3)$ in question.

19. The case just considered is that of a curve in such that through every point of the curve passes a line meeting the curve in four lines through three points; the like investigation, taking a point of the curve, gives in like manner apparently $\frac{1}{2}(m-1)(m-2)$ lines through three points; the number of these is in general $\frac{1}{6}(m-1)(m-2)(m-3)$.

20. A good illustration is afforded by taking for the curve m a curve on the hyperboloid (or quadric surface): such curves do not contain themselves as a complete intersection of the surface with another surface, but they form a fourth order surface considered as generated by a line which lies one or more times on the surface $S(m)$; and considering the curve as the generator of a quadric surface, the curve m takes only $\frac{1}{2}(q-1)^2$ times, which scrolls together make up the scroll $S(m)$; it may however happen that, e.g. of the α lines, any set or sets of α even can lie on one and the same scroll $S(m)$, that is, the scroll $S(m)$ will itself break up into scrolls of lower orders.

21. A small illustration is afforded by taking for the curve in question $S(m^2)$, such that upon a system of $S(m)$ taken $\frac{1}{2}(p)^2 + \frac{1}{2}(q)^2$ lines, and $\dots M = -pq$.

22. In the following investigations for $G(m, n, p, q)$, &c. the foregoing special cases are excluded from consideration; it may however be right to notice how it is necessary to modify the formulae in those cases.

The G formulae, Articles 23 to 34.

23. $G(m, n, p, q)$.—Considering the scroll $S(m, n, p)$ generated by a line which meets the curves m, n, p respectively, we have

$$G(m, n, p, q) = 2mnpq.$$

24. $G(m^2, n, p)$.—In a precisely similar manner we find

$$G(m^2, n, p) = np([m]_2 + M).$$

25. $G(m^3, p)$.—I investigate the value by a process similar to that employed for $G(1, 1, m)$. Suppose that the curves m and a plane curves having respectively h and k double points; then the curves m and the plane curves meet in hk points, combining with the plane curves in the following way: in question is met by the curve in $h - k$ points; the construction follows as follows:

$$G(m^3, p) = p([m]_3 + N \cdot \frac{1}{2}[m]_2 \cdot \frac{1}{2} + MN).$$

26. $G(m^4)$.—We have here

$$G(m^4) = [m]_4 + N \cdot [m]_2 \cdot \frac{1}{2} - 2[m]_2 + M^2 \cdot \frac{1}{2}.$$

27. Before going further, I observe that there are certain functional conditions which must be satisfied by the G formulae. Thus in $G(m, n, p, q)$, if q takes the form p , the formula is to reduce itself to $G(m, n, p^2)$, that is, with p^2 instead of pq ; or what is the same thing, $G(m, n, p, p) = 2G(m, n, p^2)$; and similarly $G(m, n, p, q)$ in like manner.

28. Next, before quitting the cases $G(1, 1, m)$, &c., I observe that the values of these are known directly from the consideration of an evanescent generating line, which is, in this case, simply the line of intersection of the two planes. Thus for the case $G(1, 1, m)$ we have

$$G(1, 1, m) = 2m, \quad ND(1, 1, m) = [m]_2 + M,$$

and so on.

29. Hence in $S(1, m, n)$ the formulae are now at once obtained; that is, $S(1, m, n) = 2mn$, $ND(1, m, n) = \frac{1}{2}mn(mn + m + n - 3)$.

30. First, if $G(m)$ denote $G(1, 1, m)$, and the like for $G(m^2)$ denotes $G(1, 1, m^2)$ and so on; which is the case in question; hence

$$G(1, 1, m) = 2m, \quad G(1, 1, m^2) = [m]_2 + M.$$

A solution of which is $G(m^2) = \frac{1}{2}[m]_2 + mM$. Hence $G(1, 1, m^2) = \frac{1}{2}m([m]_2 + M)$, and so we have thus

$$G(1, m^2) = \frac{1}{2}m([m]_2 + M).$$

31. Suppose then the curves m, m' are formed in the way described, viz. if $G(m, m') = (m - m')p$, and observing that $G(m, m')$ stands for $G(1, m, m')$, and that

$$G(m, m') = \frac{1}{2}m'([m]_2 + M) + \frac{1}{2}m([m']_2 + M'),$$

a particular solution is $\phi(m^2) = \frac{1}{2}[m]_2 + mM$, and we have therefore

$$\phi(m^2, m') = M(\frac{1}{2}[m]_2 + mM) + M'm + \alpha M,$$

and we have

$$NT(m^2, m') = \frac{1}{2}m'(m^2 - m) + M(m^2 - m) + \beta M.$$

32. Again, $G(m^3)$ standing for $G(1, 1, m^3)$, $G(m^3, m')$ for $G(1, m^3, m')$, the values whereof have been already found, we have

$$G(m^3, m') = m'(\frac{1}{6}[m]_3 + M(\frac{1}{2}[m]_2 + m) + M^2 \cdot \frac{1}{2}),$$

a solution of which is $G(m^3) = \frac{1}{6}[m]_3 + M(\frac{1}{2}[m]_2 + m) + M^2 \cdot \frac{1}{2}$. Now, in the same way we may obtain

$$G(1, m^3) = \frac{1}{6}m([m]_3 + M([m]_2 + m) + M^2m),$$

and the like for others.

33. Hence, substituting for $\phi(m)$, and $\phi(m, m^3)$, $\phi(m^2, m)$ their values, we have

$$\phi(m + m')^4 - \phi(m^4) - 4(\phi(m^3, m') - \phi(m, m'^3)) = mM'(m'^2 - 1) + m'M(m^2 - 1) + 6mm'MM',$$

which is satisfied. Hence the formula gives the general solution in the form

$$\phi(m^4) = M(\frac{1}{2}[m]_3 + (\frac{1}{2}[m]_2 + mM) + M^2(\frac{1}{2}[m]_2 + \frac{1}{2}M)).$$

34. Next, $G(m^2, m')$ standing for $G(1, m^2, m')$, the values whereof have been already found, we have

$$G(m^2, m') = m'(\frac{1}{2}[m]_2 + m \cdot M + \frac{1}{2}[m]_2 + \frac{1}{2}M^2 - \frac{1}{2}).$$

The S formulae, particular cases, Articles 35 to 40.

35. The S formulae have in fact been obtained in the investigation of the G formulae; we have

$$S(m, n, p) = 2mnp, \quad S(m^2, n) = n([m]_2 + M), \quad S(m^3) = \frac{1}{6}[m]_3 + M[m]_2.$$

36. In confirmation of the formula $S(1, m^2) = [m]_2 + M$, it is to be remarked that if we take through the line 1 an arbitrary plane, this meets the curve m in m points, and joining these two together by chords, each pair of which gives a generating line of the scroll, the lines on the scroll $S(1, m^2)$ which contain $[m]_2$ lines; that is, the order of the surface $S(1, m^2)$ is $[m]_2 + M$.

37. In confirmation of the formula $S(1, m^2) = [m]_2 + M$, it is to be remarked that if we take through the line 1 a plane, this plane meets the curve m in m points; and joining these two together by chords, each pair of which gives a generating line of the scroll, the lines on the scroll $S(1, m^2)$ in the plane meet in m points, that is, in the order of the scroll $S(1, m^2) = [m]_2 + M$.

38. As regards the formula $S(m^3) = \frac{1}{6}[m]_3 + M[m]_2$, suppose that the curve m is on a quadric surface; then there cannot in general be a line meeting it in three points; that is, the order of the scroll $S(m^3) = 0$ in the case considered. But if it should happen that the chord of the curve meets it in three points, the construction would still be valid; in the present case $M = -pq$; whence in this case the order of the scroll $S(m^3)$ is calculated as

$$S(m^3) = \frac{1}{6}[m]_3 + M[m]_2 = \frac{1}{6}pq(2pq - p - q - 3)(2pq - p - q - 2);$$

which on putting therein $p + q = m$, $pq = \beta$ and from the reduced expressions obtaining the values of $S\beta$, &c., give

$$\Sigma\alpha = \beta - \frac{1}{4}\alpha^2 + \frac{1}{2}\alpha - 18,$$

$$\Sigma\alpha' = \beta - (-\alpha) + \frac{1}{2}\alpha^2 + \frac{1}{4}\alpha - 58,$$

$$\Sigma\alpha^2 = \beta^2 + 2\alpha\beta + \frac{1}{2}\beta^2 - \frac{1}{4}\beta\alpha + \frac{1}{4}\alpha^2 + \frac{4}{3}\alpha^3 - 119\alpha + 198,$$

$$\Sigma\alpha\beta = \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2}\alpha + \frac{1}{2}\alpha^2 - \frac{1}{2}\alpha) + \frac{1}{2}\alpha^3 + \frac{4}{3}\alpha^2 - \frac{11}{12}\alpha + 19,$$

$$\Sigma\alpha\beta\gamma = \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2} - \alpha + \frac{1}{2}\alpha^2) + \beta(\frac{1}{4}\alpha^4 - \frac{1}{2}\alpha^3 + \frac{4}{3}\alpha^2 - \frac{11}{12}\alpha + 19),$$

$$\Sigma\alpha\beta\gamma\delta = \beta^3(\frac{1}{2}) + \beta^2(-\frac{1}{6}\alpha^2 + \frac{1}{2}\alpha - \frac{4}{3}) + \beta(\frac{1}{2}\alpha^4 - \frac{1}{4}\alpha^3 + \frac{19}{4}\alpha^2 - \frac{1067}{96}\alpha + 1071\alpha - 1560);$$

this, with the reduced expressions for $S(m^3)$, gives the final form.

39. We have

$$\begin{aligned}\Sigma\alpha &= -\beta - \frac{1}{4}\alpha^2 + \frac{1}{2}\alpha - 18, \\ \Sigma\alpha' &= -\beta - \frac{1}{4}\alpha^2 + 9\alpha - \frac{1}{2}\alpha^3 + 38, \\ \Sigma\alpha^2 &= \beta^2 + 2\alpha\beta + \frac{4}{3}\alpha^4 + 9\alpha^3 - \frac{11}{4}\alpha^3 + \frac{1}{4}\alpha^4 + 90\alpha + 198,\end{aligned}$$

and so we have for the order

$$\Sigma B = \frac{1}{6}pq(\alpha^2 - 5\alpha + 6) + Mn(2p - q - 1) - 119\alpha + 198,$$

where, multiplying $\alpha^2 - 5\alpha + 6 = 0$ gives $\alpha = 2$ or $\alpha = 3$, and the formula gives the general solution.

40. In the same manner the order of the scroll $S(m^3)$ is obtained, and combining the previous expressions, we find the final formula for the order of the surface $S(m^3)$:

$$S(m^3) = \frac{1}{6}m(m-1)(m-2) + M(m-1)(m-2),$$

together with such other contributions as arise from the special cases when the curve m has multiple points or other singularities.

Intersections of a generating line with the Nodal Total, Articles 59 to 63.

59. We may for the scrolls $S(1, m, n)$ and $S(1, m, n^2)$ verify the theorem that each generating line meets the Nodal Total in a number of points $= S - 2$.

In fact for the scroll $S(1, m, n)$ the directrix curves are respectively multiple curves of the orders m, n , and a generating line meets each of these in a single point. The number of intersections therefore is a single $24 - 2$

60. Similarly for the scroll $S(1, m, n^2)$ the directrix curves are multiple curves, viz. the curve 1 is a $(\frac{1}{2}n(n-1) + M)$ -tuple curve; the generating line meets 1 in a single point, contained on 1 in a $\frac{1}{2}n(n-1) + M$ -tuple point; thus the curve 1 is met in a single point, contained on 1 in a $\frac{1}{2}n(n-1) + M$ -tuple point; the directrix is therefore $\frac{1}{2}n(n-1) + M$ points, each of which counts simply, and the latter in $(n-2)$ points, each of which counts as a single point; and the value is $1 \cdot \frac{1}{2}n(n-1) + M + \frac{1}{2}(n-1)(n-2)(n-3)$.

61. For the scroll $S(m, n, p)$ the directrix is met in a single point, contained on 1 in a $\frac{1}{2}n(n-1) + M$ -tuple point; the directrix is therefore $\frac{1}{2}n(n-1) + M$ points, each of which counts simply, and the value is

$$S(m, n, p) - 2 = 2mnp - 2.$$

62. For the scroll $S(m^2, n)$ we find similarly

$$S(m^2, n) - 2 = n([m]_2 + M) - 2.$$

63. And lastly for the scroll $S(m^3)$ we have

$$S(m^3) - 2 = \frac{1}{6}m(m-1)(m-2) + M(m-1)(m-2) - 2,$$

which gives

$$S(m^3) = \frac{1}{6}m(m-1)(m-2) + M(m-1)(m-2).$$

The foregoing expressions for the scroll $S(m^3)$ might with propriety have been inserted in the Table.

Annex No. 1.—Investigation of the formula for $S(m^2)$ in the case of the unicursal curve (referred to, Art. 39).

Consider the unicursal m -thic curve the equations whereof are

$$x : y : z : w = A : B : C : D,$$

where A, B, C, D are rational and integral functions of a parameter θ ; and let it be required to find the equation of a plane meeting the curve in such manner that three of the points of intersection are in lined. Taking the points $(\xi, \eta, \zeta, \omega)$ as $(\theta_0), (\theta_1), (\theta_2)$ on the curve, we find between $(\xi, \eta, \zeta, \omega)$ and $(\xi', \eta', \zeta', \omega')$ an equation of a certain degree in $(\xi, \eta, \zeta, \omega)$ and $(\xi', \eta', \zeta', \omega')$, which is the equation of the plane meeting the curve at three points

$$\xi x + \eta y + \zeta z + \omega = 0,$$

we find between $(\xi, \eta, \zeta, \omega)$ an equation of a certain degree in θ ; and to express that three of the roots be equal, we have between $(\xi, \eta, \zeta, \omega)$ a system of equations, which I do form the equations

$$\begin{vmatrix} \lambda, \mu, \nu, \rho \\ A_0, B_0, C_0, D_0 \\ A_1, B_1, C_1, D_1 \\ A_2, B_2, C_2, D_2 \end{vmatrix} = 0;$$

and if we form the equation

$$\Pi \begin{vmatrix} \lambda, \mu, \nu, \rho \\ A_0, B_0, C_0, D_0 \\ A_1, B_1, C_1, D_1 \\ A_2, B_2, C_2, D_2 \end{vmatrix} = 0,$$

where Π denotes the product of the terms belonging to all the triads of the m roots, the result will be symmetrical in regard to all the roots; and replacing the symmetrical functions of the roots by expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}m(m-1)(m-2)} = 0,$$

or, the symmetrical functions are expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]_3} = 0.$$

Now the above-mentioned determinant is divisible by $(\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2)$; for it vanishes when any two of the indeterminates $\theta_0, \theta_1, \theta_2$ become equal; and the same product divides Π . Considering only the residue, we have

$$\Pi (\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2) = \text{Disct.}(\xi, \eta, \zeta, \omega)^{[m]_3},$$

or, say

$$\Pi (\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2) \text{ is the Discriminant in respect to } (\xi, \eta, \zeta, \omega)^{[m]_3}.$$

Now the above-mentioned determinant is divisible by $(\theta_0 - \theta_1)(\theta_0 - \theta_2)(\theta_1 - \theta_2)$, and the symmetrical functions are expressed in terms of the coefficients; thus the formula

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]_3}$$

contains linearly the coefficients of the determinant, that is, of (A_i, B_i, C_i, D_i) of the orders θ_i ; that is, the form is symmetric in regard to all the roots; and replacing the symmetrical functions of the roots by their values expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]_3}.$$

The like remarks apply to the lines $G(m^2, n, p)$, $G(m^2, n)$, and $G(m^3)$; but I will develop them somewhat more particularly as regards the line $G(m^3)$ in question.

Annex No. 2.—Investigation of the formula for $S(m^2)$, when the curve m is the pq complete intersection, viz. when it is the intersection of two surfaces of the orders p and q respectively (referred to, Art. 40).

Let $U = 0$, $V = 0$ be the equations of the two surfaces of the orders p and q respectively. Take (x, y, z, w) the coordinates of a point on the curve, and the equations of the two curves respectively, write for the coordinates $x + px'$, $y + py'$, $z + pz'$, $w + pw'$; then putting

$$\Delta = x\partial_x + y\partial_y + z\partial_z + w\partial_w,$$

the resulting equations may be represented by

$$(\Delta U, \Delta^2 U, \dots \Delta^p U \text{ } \S 1, \rho)^{p-1} = 0, \quad (\Delta V, \Delta^2 V, \dots \Delta^q V \text{ } \S 1, \rho)^{q-1} = 0,$$

where it is to be noticed that besides the expressed literal coefficients there are numerical coefficients (not as the notation usually denotes, the binomial coefficients, but) $= [\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3 \cdot 2}, \frac{1}{2 \cdot 1}]$, &c.

Supposing that (x', y', z', w') are the current coordinates of a point on the line drawn through the point (x, y, z, w) to meet the curve m in two other points, and equations in ρ must have two common roots, and to express this we have a system of two other equations, viz. if $k = \frac{1}{2}(p + q - 1) + \frac{1}{2}(q - 2)$; that is, $k = \frac{1}{2}(p + q)$.

2. The use of the combinations $(\Delta U, \Delta^2 U, \dots \Delta^p U)$, $(\Delta V, \dots)$ are all in either group in the Appendix; if $k = \frac{1}{2}[p]_2 + \frac{1}{2}[q]_2$; that is, k in either case is in the Appendix and his Treatise on the Analytic Geometry of Three Dimensions; and in the investigation in the following Annexes 2 and 3, the route which I have followed are completely traced out for me by him, in this way which I most readily presented itself.

Annex No. 2.—Investigation of the formula for $S(m^2)$ when the curve m is the complete intersection, viz. when it is the intersection of two surfaces of the orders p and q respectively (referred to, Art. 40).

Let $U = 0$, $V = 0$ be the equations of the two surfaces of the orders p and q respectively. Take (x, y, z, w) the coordinates of a point on the curve, the equations of the two curves respectively, write for the coordinates $x + \rho x'$, $y + \rho y'$, $z + \rho z'$, $w + \rho w'$; and so on in other cases; and considering first the equation

$$(\Delta U, \Delta^2 U, \dots \Delta^p U \text{ } \S 1, \rho)^{p-1} = 0$$

of the degree p in ρ , the coordinates of a point on the curve, the equations to determine the equation of the plexus in the square +1 system, the resulting equations may be represented by:

$$\begin{vmatrix} \Delta U, \Delta^2 U, \dots \\ \Delta U, \dots \\ \vdots \\ \Delta V, \Delta^2 V, \dots \\ \Delta V, \dots \end{vmatrix} = 0,$$

$p + q - 3$ columns, $(q - 2) + (p - q - 4)$ lines; or representing the terms according to their order and weight, that is, degree in (x, y, z, w) and (x', y', z', w') respectively (the order and weight of

the evanescent terms being fixed as in this case) they may form a regular series with the other terms, the system is

$$\left| \begin{array}{c} (p+q-3 \text{ columns}) \\ (p-1)\lambda, (p-2)\lambda_1, \dots, (p-1)\lambda_{p-2} \\ p_0, p_1, \dots, p_{p-3} \\ \vdots \\ (q-1)\lambda_0, (q-2)\lambda_1, \dots, (q-1)\lambda_{p-2} \\ q_0, q_1, \dots, q_{p-3} \end{array} \right| = 0,$$

so that

$$\begin{aligned} \alpha, \beta, \dots &= p-1, p, p+1, \dots, -(q-2), 1, 0, -1, \dots, -(p-3), \\ \alpha', \beta', \dots &= 1, 0, \dots, -q+4, \quad q, q-1, q-2, \dots, p+q-5, \\ A, B, \dots &= -1, -2, \dots, p+q-4, \dots, -(p+q-4), \\ A', B', \dots &= 1, 2, \dots, \dots, p+q-4. \end{aligned}$$

We then find

$$\begin{aligned} \Sigma\alpha &= (q-3)(p-2) + \frac{1}{2}(q-2)(q-1) + (p-2)\frac{1}{2}a + 10, \\ \Sigma\alpha' &= -\beta - \frac{1}{4}a^2 + \frac{1}{2}a - 10, \\ \Sigma A &= -\Sigma A' = \frac{1}{2}a^2 - \frac{1}{2}a + 6, \\ \Sigma a^2 &= \frac{1}{8}a^3 - \frac{1}{4}a^2 + \frac{14}{3}a + 26, \\ \Sigma aa' &= -\Sigma AA' = -\frac{1}{3}a^3 - \frac{1}{4}a^2 + \frac{1}{4}a - 90a + 100, \\ \Sigma AB &= \frac{1}{8}a^3 - \frac{19}{12}a^2 + \frac{19}{4}a^3 - \frac{47}{12}a + 65, \\ \Sigma ABC &= \frac{1}{120}a^4 - \frac{17}{24}a^3 + \frac{31}{8}a^2 - \frac{1379}{120}a + 19. \end{aligned}$$

We find

$$\begin{aligned} \Sigma A + \Sigma a &= \beta - 8, \\ \Sigma A^2 + \Sigma a^2 &= \beta - 3a + 8, \\ \Sigma AB - \Sigma a\beta &= \beta^2 - \beta(-\frac{1}{2}a + \frac{1}{2}\beta - \frac{4}{3})a + 36a - 126, \\ \Sigma AB - 2a\beta &= \beta(-\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + \frac{4}{3}a + \frac{1}{4}) + (-\frac{1}{12}a^4 + \frac{4}{3}a^2 - \frac{47}{12}a + 16a) - 4a^4 - 36a + 144, \\ (\Sigma AB - \Sigma a\beta) + 2a(\Sigma A + \Sigma a) &= \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + \frac{4}{3}a + 144), \\ \frac{1}{2}(\Sigma AB + \Sigma a\beta + 2a(\Sigma A + \Sigma a)) &= \beta(\frac{1}{2}a) + \beta(-15a + 28) + 72a - 228, \end{aligned}$$

and

$$\Sigma A^2 A' + \Sigma a^2 a' = (\text{ut suprà}) \beta^2 + (-\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + 9a - \frac{4}{3}) - 29a + 98;$$

whence, adding these three expressions, we find

$$\text{Weight} - \text{Order} = \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + \frac{4}{3}a + \frac{4}{3}) + \beta(-\frac{1}{12}a^4 + \frac{2}{3}a^3 - 7a^2 + \frac{77}{6}a - 18);$$

and for the order we have

$$(\Sigma a)^2 - \Sigma a\beta = \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + \frac{4}{3}a) + \beta(\frac{1}{2}a^3 + 3a + \frac{1}{4}\beta + 133);$$

and then

$$\begin{aligned} \Sigma ABC + \Sigma a\beta\gamma &= (\text{ut suprà}) \beta(-\frac{1}{12}) + \beta(\frac{1}{12}a^2 - \frac{1}{4}a + \frac{1349}{120}) - 1210, \\ (\Sigma AB - \Sigma a\beta)(\Sigma A + \Sigma a) &= \beta^2(-\frac{1}{4}a) + \beta(\frac{4}{3}a^2 + \frac{49}{4}a^2 - \frac{14}{12}a + 241), \end{aligned}$$

$$-a^4 + 12a^3 - 132a + 1064;$$

whence, adding these three expressions, we find for the weight

$$\text{Order} - \text{Order} = \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + \frac{4}{3}a + \frac{4}{3}) + \beta(\frac{4}{3}a^3 + 3a - \frac{4}{3}a) - 6;$$

and by the foregoing expression for the weight, we get

$$\text{Weight} - \text{Order} = \beta^2(\frac{1}{2}) + \beta(-\frac{1}{2}a^2 + \frac{4}{3}a + \frac{4}{3}) - 11a + 24;$$

and therefore

$$G(m^2) = \frac{1}{2}\beta[2\beta^2 + \beta(-6a) + (3a^2 + 18a - 26) - 66a + 144],$$

a formulae the direct verification whereof is due to Dr Salmon; see *post*, Annex No. 3.

The formulae for $NT(1, m, n)$ and $NR(1, m, n)$, Articles 43 to 46.

43. $NR(1, m, n)$.—Through the line 1 take any plane meeting the curve m in m points and the curve n in n points; in this case, by the construction of the scroll, we have through each of the m points, lines n in number forming the lines of the scroll $S(1, m, n)$, and their intersections are the points $m_1n_1, m_2n_2, \dots, m_mn_m$, are generating lines of the scroll; their intersection on the line 1 in question being a point of the line 1 in the scroll, and on the line 1 being a single point of the line 1. Through each of the n points, that is, the lines 1 being on the surface, they are generating lines of the scroll.

44. The points (a) are included among the cuspidal points on the line 1. Taking for a moment $x = 0, y = 0$ for the equations of the line 1 (which, as we have seen, are a non-singular line on the scroll), we have a system of intersections with the curves m and n giving rise to such a relation as a, a' being functions of the coordinates, the degree in (x, y, z, w) are then to make a particular system of values

$$2[mn]_2 = 2n + 2x' + R,$$

see *post*, Annex No. 4. And then observing that the form

$$x = S(m, n^2) = mn([n]_2 + N),$$

these values give

$$\begin{aligned} 2x + 2x' + R &= 2n(mn^2 - 2x - 2x' - R), \\ &= [mn]_2 - n[m]_2 - x[n]_2, \\ &= [mn]_2; \end{aligned}$$

and thence

$$NR(1, m, n) = \frac{1}{2}mn(mn - m - n - 3),$$

which is the result before noted in the Table.

45. $NR(1, m^2)$.—Through any line 1 take any arbitrary plane meeting the curve m in m points; if $m_1, m_2, \dots, m_m$ are the points of intersection of the curve, the lines so determined are generating lines of the scroll $S(1, m^2)$, and their intersections on the line 1 in m points; and their intersections in the plane 1 in m points yielding the nodal cuspidal points NR ; but in like manner the lines $m_1, m_2, m_3, m_4, m_5, m_6$ are generating lines of the scroll of an arbitrary plane through the curve m in m points, that there are also generating lines of the scroll of an arbitrary plane through the curve m in m points. We have therefore that

$$NR(1, m^2) = \frac{1}{2}m([m]_2 + M) + a.$$

But these formulae obtained in fact give them as the special cases. To find them as the special cases, we will employ

$$S(m, n) = nS(1, m, n) + ND(1, m, n) = 2mn,$$

that is, $S = S(1, 1, n) + 2[m]_2$. Hence ultimately

$$S = 2m,$$

since 2 is the number of lines which can be drawn meeting each of four given right lines. Hence

$$S(1, 1, 1, 1) = 0 \quad (m = 1, n = 1, p = 1), \quad \text{that is,} \quad S(1, 1, 1, 1) = 2.$$

46. $G(m, n, p, q) = 2mnpq$. — We have here a precisely similar manner to

$$G(m, n, p, q) = 2mnpq.$$

Intersections of a generating line with the Nodal Total, Articles 47 to 58.

47. We have

$$\begin{aligned} NT(1, m, n) &= NG(1, m, n) + ND(1, m, n) - mn(m + n - 2) + ND(1, m, n) + NR(1, m, n) \\ &\quad + NR(1, m, n) = \frac{1}{2}mn(mn + 3m + 3n - 5), \end{aligned}$$

which is the result in the Table.

48. And moreover

$$\begin{aligned} NT(m^2, n) &= ND(m^2, n) + NG(m^2, n) - ND(1, m, n) \\ &\quad + NR(m^2, n) = \frac{1}{2}[m]_2 + \frac{1}{2}[m]_2 + M([m]_2 - \frac{1}{2}) + M^2 \cdot \frac{1}{2} \\ &\quad + NR(m^2, n) + 2[m]_2 + M([m]_2 - \frac{1}{2}) + M^2 \cdot \frac{1}{2} + M([m]_2 - 2m + 5); \end{aligned}$$

which is

$$\begin{aligned} &= \frac{1}{2}[m]_2 + 2[m]_2 + M([m]_2 - \frac{1}{2}) + M^2 \cdot \frac{1}{2}, \\ &= \frac{1}{2}[m]_2 - S + M([m]_2 - \frac{1}{2}), \end{aligned}$$

if $S = S(1, m^2) = [m]_2 + M$.

The NT and NR formulae, Articles 49 to 58.

49. I proceed to find $NT(m, n, p)$, &c. by a functional investigation, such as was employed for finding $G(1, 1, m)$, &c. Writing $S(m)$ to denote either of the scrolls $S(m, n, p)$ or $S(m^2, n)$ or $S(m^3)$, the aggregate of the two curves m, n' ; then the curve m' may be replaced by h a double point and join two together by chords, each pair of which gives a generating line of the scroll; the latter in question are merely a double curve, viz. a chord of the line $G(m, m')$. We have

$$NT(m + m')^4 = NT(m^4) + NT(m'^4) + 6(m + m'^4) \left(\frac{1}{2}[m]_3 + M([m]_2 + m) + M^2(\frac{1}{2}m + 1) + MM' \right).$$

50. Instead of assuming

$$NT = \frac{1}{2}S^2 + \phi,$$

it is the same thing, and it is rather more convenient, to assume

$$NT = \frac{1}{2} S^2 - S + \phi,$$

viz. $NT - \frac{1}{2} (S(m))^2 - S(m) + \phi(m)$, &c. Then observing that

$$S(m + m') = S(m) + S(m'), \text{ \&c.}$$

51. First, if $\phi(m)$ stands for $\phi(m, n)$ or $\phi(m^2)$ or $\phi(m^3)$, the foregoing equations give

$$\phi(m + m') = \phi(m) + \phi(m'),$$

$$\phi(m + m^2)^3 = \phi(m^3) + \phi(m'^3) + 3\phi(m^2, m') + 3\phi(m, m'^2),$$

$$\phi(m + m')^4 = \phi(m^4) + \phi(m'^4) + 4\phi(m^3, m') + 4\phi(m, m'^3) + 6\phi(m^2, m'^2);$$

and if in the second equation $\phi(m, m')$, $\phi(m^2, m')$ and in the third equation $\phi(m^2)$, $\phi(m^3)$ are regarded as known, these are all of them of the form

$$f(m + m') - f(m) - f(m') = \text{Funct. } (m, m'),$$

so that, a particular solution being obtained, we have at least $f(m)$ as a function of m and M only.

52. First if $\phi(m)$ stands for $\phi(m, n, p)$, we obtain $\phi(m, n, p) = mn + \beta M$, or observing that $\phi(m, n, p)$ must be symmetrical in regard to the curves m , n , and p , we may write

$$f(m, n, p) = mnp + \beta(Mnp + Nmp + Pmn) + \gamma(mNP + nMP + pMN) + \delta MNP,$$

and then

$$\begin{aligned} NT(m, n, p) &= \frac{1}{2} S^2 - S + \phi(m, n, p), \\ &= 2mnp(mnp - 1) + \phi(m, n, p). \end{aligned}$$

A particular solution is $\phi(m, n, p) = (m^2)N + mnp$, and thus we have therefore

$$\phi(m^2, n) = M(\frac{1}{2}[m]_2 + mn) + M^2 n + Mn + \beta M,$$

or observing that n , considered as a function of n only, satisfies the equation

$$\phi(n + n') = \phi(n) + \phi(n'),$$

we then have

$$NT(m^2, n) = \frac{1}{2} S^2 - S + \phi(m^2, n),$$

where

$$S = S(m^2, n) = n([m]_2 + M).$$

53. Next, if $\phi(m^2)$, substituting for $\phi(m, n)$ and $\phi(m, n')$ their values, we have

$$\phi(m + m')^4 - \phi(m^4) - \phi(m'^4) = m([m]_2 + M) + m'([m']_2 + M') + mn'(m + m')$$

which is satisfied by

$$\phi(m^4) = M(\frac{1}{2}[m]_3 - \frac{1}{2}m) + M^2 + am,$$

and the general value then is

$$NT(m^4) = M(\frac{1}{2}[m]_3 - \frac{1}{2}m) + M^2 + am + \beta M,$$

and we have

$$NT(m^4) = \frac{1}{2} S^2 - S + \phi(m^4).$$

54. Taking for the curve in question (p, q) curve on the hyperboloid $(m = p + q, M = -pq)$, $S(m^4)$ becomes the hyperboloid taken k times, if $k = \frac{1}{2}[p]_2 + \frac{1}{2}[q]_2$; that is, the nodal curve is the hyperboloid taken k times, the values NT and S that imply $a = 2, \beta = 11$, for the values which the formula gives

$$\phi(m^4) = \frac{1}{2} m(m^2)(-7) + mM + p(m-1)Mm(m-1).$$

55. I assume the correctness of the value

$$\phi(m^4) = 2m + M(\frac{1}{2}[m]_3 - \frac{1}{2}m) + M^2;$$

so obtained, as being in fact verified by means of the six several curves $(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)$, &c. in the foregoing value of $\phi(m^4)$, we have therefore the formula

$$\phi(m^4) = 2m + M(\frac{1}{2}[m]_3 - \frac{1}{2}m) + M^2,$$

and so on; thus the value of $\phi(m^4)$ thus obtained is being in fact verified by means of the six several curves $(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)$, &c. in the foregoing value of $\phi(m^4)$.

56. The points (a) are included among the cuspidal points on the line 1. Taking for a moment $x = 0, y = 0$ for the equations of the line 1 (which, as we have seen, is a non-singular line on the scroll), we have a relation as a, a' being functions of the coordinates, the degree in (x, y, z, w) being a system of values, $a = a', b = b'$, where the lines 1 are in the same plane.

57. For the scroll $S(m, n^2)$ we find

$$\begin{aligned} NR(m, n^2) &= NT(m, n^2) - ND(m, n^2) - NG(m, n^2), \\ &= a(\frac{1}{2}[n]_2 + N(\frac{1}{2}[n]_2 + n) + N^2 \cdot \frac{1}{2}m). \end{aligned}$$

58. And moreover

$$\begin{aligned} NR(m^3) &= NT(m^3) - ND(m^3) - NG(m^3), \\ &= m(\frac{1}{2}[m]_2 + \frac{1}{2}[m]_2 + M([m]_2 - \frac{1}{2}) + M^2 \cdot \frac{1}{2}M) \\ &\quad + M(\frac{1}{2}[m]_2 + \frac{1}{2}[m]_2 + 9m - 20) + M^2 \cdot (\frac{1}{2}[m]_2 - 2m). \end{aligned}$$

and the investigation of the series of results given in the Table is thus concluded.

Intersections of a generating line with the Nodal Total, Articles 59 to 63.

59. We may for the scrolls $S(1, m, n)$ and $S(1, m, n^2)$ verify the theorem that each generating line meets the Nodal Total in a number of points $= S - 2$.

In fact for the scroll $S(1, m, n)$ the directrix curves are respectively multiple curves of the orders m, n , and a generating line meets each of these in a single point. The number of intersections is therefore

$$m + n - 2 = S(1, m, n) - 2.$$

60. Similarly, counting in like manner the lines m, n , and considering the construction of the surface, we will obtain in addition the value a of the points $(\alpha, \beta, \gamma, \delta)$ on the curve and

considering the union of the nodal director, $G(m^2, n)$ being equivalent to $\frac{1}{2}mM$, we find that the curve m_1, m_2 are not the same plane a, b , those the lines n_1, n_2, n_3 , are generating lines of the scroll, that there are, e.g. of the α lines, α in number; whence in question does not arise of finding the number of additional generating lines on the scroll.

61. The torse case considered is that of a curve in space, that is, a curve in space, but it is the curve which lies on a quadric surface, and is the complete intersection of a quadric surface with a surface of the order p or q respectively; but I will developpe them somewhat more particularly as regards the discussion of the residue, the order of the residue is in question; the number of points in space considered as the residue, the order of the residue is in question.

62. The values of $G(m, n, p, q)$, $G(m^2, n, p)$, $G(m^3, p)$, $G(m^4)$ are subjected to the verification which has been already mentioned.

63. We see now that the values S , ND , NG , NR , NT for each of the scrolls of the form $S(m, n, p)$, $S(m^2, n)$, and $S(m^3)$ are all those which have been already given in the Table.

[End of Papers 334 (cont.), 336–339, from pp. 151–200 of Vol. V]